

230. Kth Smallest Element in BST

PLATFORM

Platform: LeetCode

DIFFICULTY

MEDIUM

DATE

Solved: August 3, 2021

Problem Summary

Given the root of a BST and an integer k , return the k th smallest value among all node values.

Approach

In-order traversal of BST gives sorted ascending order. Maintain a global counter — when counter hits k , store current node's value and stop.

Algo

1. Initialize count = 0, result = 0.
2. In inOrder: if root == null return.
3. Recurse left.
4. count++ — if count == $k \rightarrow$ store result = root.val and return.
5. Recurse right.
6. Return result.

Key Insights

- In-order of BST = sorted ascending — k th element in traversal = k th smallest.
- Early termination: stops visiting as soon as k th element is found — doesn't traverse entire tree.
- Global count and result track state across recursive calls.

Time Complexity & Space Complexity

$O(h + k)$ — traverse h levels to reach leftmost, then k steps.

$O(h)$ — recursion stack.

Observations

- Without early termination this is just LC 94 (in-order) + array lookup — always add early termination.
- For frequent k th queries with many insertions, augment BST with subtree sizes — $O(\log n)$ per query.
- return after storing result doesn't fully stop recursion — right subtree calls still happen but count $> k$ so nothing changes.

Approaches

- **Brute Force $O(n)$ space:** Run full in-order traversal, store all values in a list, return list.get($k-1$).
- **Optimal $O(h)$ space:** Run in-order recursively with a counter, stop as soon as count reaches k — never visits unnecessary nodes.

Problem List

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Hint

Given the `root` of a binary search tree, and an integer `k`, return the k^{th} smallest value (1-indexed) of all the values of the nodes in the tree.

Example 1:

```
graph TD; 3((3)) --- 1((1)); 3 --- 4((4)); 1 --- 2((2))
```

Input: `root = [3,1,4,null,2]`, `k = 1`
Output: 1

Example 2:

```
graph TD; 5((5)) --- 3((3)); 5 --- 6((6)); 3 --- 2((2)); 3 --- 4((4))
```

Accepted

93 / 93 testcases passed

Sahil Khan submitted at Aug 03, 2026 11:00

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Runtime

0 ms | Beats 100.00%

Memory

46.22 MB | Beats 97.72%

Interval	Performance
1ms	High
2ms	Low
3ms	Low
4ms	Low

Code

Java

```
1  * }
2  */
3  class Solution {
4      int KthSmallest = 0;
5      int count = 0;
6
7      public int kthSmallest(TreeNode root, int k) {
8          solve(root, k);
9          return KthSmallest;
10     }
11
12     void solve(TreeNode root, int k){
13         if(root == null) return ;
14
15         solve(root.left, k);
16         count++;
17         if(count == k){
18             KthSmallest = root.val;
19             return;
20         }
21         solve(root.right, k);
22     }
23 }
```

Testcase

Test Result

Accepted

Runtime: 0 ms

Case 1

Case 2

Input

root = [3,1,4,null,2]

k =

12.9K

226

102 Online